FS Engine Integration — Instruction Prompt

For Google AI Studio Build mode · FS Prep project

Version 1.0.0 · 2026-10-21 | Paste the prompt below into AI Studio with the two attachments in place.

Attachments that go with this prompt

- **FS_Engine_Integration_Spec_v1.0.0.pdf** — the full specification. It is the source of truth; where this prompt and the spec ever differ, the spec wins.
- **bank-seed-20261021.json** — the question bank data file (480 questions, engine schema v1). Attach the raw .json file — never a PDF or screenshot of it.

The prompt (copy everything in the box)

Connect this app to its question engine, following the attached specification exactly: “FS Engine Integration Spec v1.0.0” (PDF). A question bank file, bank-seed-20261021.json, is also attached.

Background in one paragraph: this app currently hardcodes 32 practice questions in QuestionBank.kt. Those questions actually belong to a separate engine that generates large question banks as JSON files in a locked format (“schema v1”). Your job is to delete the hardcoded questions and make the app load the attached engine bank instead. This is a data-layer change only: do not change any screen, navigation, theming, or feature behavior beyond the specific items listed below.

1. Place the bank file. Put the attached bank-seed-20261021.json at app/src/main/assets/banks/bank-seed-20261021.json, byte-for-byte unchanged.

2. Build a bank loader. At startup, load and parse that asset once, using only org.json (built into Android). Do NOT add any JSON library or any other new dependency. The file shape is: a top-level object with “engine_version”, “generator”, “question_count”, and “questions” (an array). Each question object has: qid, schema_version, domain (a slug string), topic, difficulty (integer 1–3), stem, choices (array of 4 strings), answer_index (integer 0–3), explanation, parameters (object — ignore its contents), seed (integer — ignore).

3. Map to the existing Question model exactly as the spec's mapping table says: qid → id; stem → questionText; choices → options; answer_index → correctIndex; explanation → explanation; topic → topicTag; difficulty 1 → EASIER, 2 → ABOUT_THE_SAME, 3 → HARDER; formulaHighlight → null.

4. Validate while loading, and fail visibly. If any of these checks fail, show an error screen/message naming the specific problem and load NO questions — never fall back to any built-in or sample question set: (a) the bank's schema_version is not 1; (b) question_count does not equal the actual number of questions; (c) any qid is empty or duplicated; (d) any question does not have exactly 4 unique, non-empty choices; (e) any answer_index is outside 0–3; (f) any stem or explanation is empty.

5. Replace the domain taxonomy. Replace the FsDomain enum with exactly these 7 entries (slug / title / short code) and a one-line description each: applied-math-statistics / “Applied Math & Statistics” / MATH · boundary-law-and-real-property / “Boundary Law & Real Property” / LAW · business-concepts-ethics / “Business Concepts & Ethics” / ETHICS · mapping-processes-and-methods / “Mapping Processes & Methods” / MAP · survey-computations / “Survey Computations” / COMP · surveying-principles-geodesy / “Surveying Principles & Geodesy” / GEOD · surveying-processes-and-methods / “Surveying Processes & Methods” / PROC. Questions select their domain by matching the slug. Remove the old targetExamQuestions field completely.

6. Keep the data API identical. The QuestionBank object must keep its existing public functions — questions, getQuestionsForQuiz, getAdaptiveQuestions, getMockExamQuestions — with the same signatures and selection semantics, now drawing from the loaded bank. Delete the entire hardcoded 32-question list. Do not change the ViewModel or any screen to accommodate this; the screens should not be able to tell the data source changed.

7. Make the mock exam real. getMockExamQuestions defaults to 110 questions drawn proportionally across the 7 domains, and the mock exam time limit becomes 320 minutes.

Explicitly out of scope — do not touch: any screen layout or navigation, the Calm Pro theme and tokens, the launcher icon, the study guide, flashcards, achievements, the leaderboard, scoring logic (grading remains selectedIndex == correctIndex), and user progress/state handling.

After the run

- Re-sync the project to GitHub.
- Muse audits the synced code against §7 of the spec and reports pass/fail per item before anything is trusted.
- Then the GUI phase: 4-tab simplification, then removal of the “NCEES” name from user-facing text.